

Gold Export Trends and Opportunities in Ghana and South Africa

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I. INTRODUCTION

Gold has been mined, formed and traded since at least 4500 BC, as archaeological discoveries show in Wana, Bulgaria, ancient Egypt and written in the Bible. Ancient societies developed coins and standardized gold ingots for trading, establishing the basis for gold as a worldwide currency. In the 17th century, gold standard as a concept appeared, linking coins' values to its gold reserves. Even though it was internationally accepted at the end of the 19th century, its implementation raised debates and economic tensions. In 1919, Rothschild bank in London started setting daily prices for gold, a practice that still remains to this day.

After World War II, the Bretton Wood Agreement from 1944 stipulated fixed exchange rates for all countries of the world based on gold. However, this system ended in the 1970s, as coins stopped being directly linked to gold. Despite this, gold still remains a key for global economies, with central banks and private stakeholders maintaining significant amounts of the said mineral. As a mercantilist, gold is a popular investment due to its stable price during the economic crisis. Nevertheless, gold is still remains subject to market fluctuations, making it attractive for speculators.

II. IMPACT OF THE COUNTRIES' RELATIONSHIPS ON GOLD TRADE

The geopolitical risks, like international political instability, have a significant effect on the management of the resources of each country. Recent examples such as COVID-19 pandemic or the Russian war have challenged the management of said resources. In order to guarantee economic stability and meet their obligations as regards external debt, countries maintain reserve portfolios. Analyzing these risks is crucial given their impact on global economic growth.

A. Main gold importers and producers

- China: in 2023 it was the biggest gold producer, with 370 metric tons (MT), maintaining the lead despite a decline from its 2016 peak of 455 MT.
- United States: in 2023 they produced 170 MT, continuing a downward trend since 2017 (237 MT). Nevada and Alaska account for 73% and 13% of production, respectively.
- United Kingdom: the fourth-largest gold importer in 2022 (\$38.7B). Gold has been part of its reserves since the 18th century, with current reserves at approximately 310 MT.
- BRICS: This block of countries focuses on sustainable development and is exploring a bank to finance infrastructure and growth.

International relations and geopolitical events have a deep impact on commerce and gold reserves. Financial stability and the reaction of the countries depend on political strategies and an adequate management of resources.

B. COVID-19's major economic impacts worldwide

The recent pandemic had big negative effects on global economy. In 2020, global GDP fell by 3.4%, representing a loss of more than two billion dollars. However, the economy recovered fast, reaching a positive growth in 2021 with a GDP of 93.6 billion dollars, and continued growing until 2022 despite the challenges originated by the war in Ukraine. Global stock markets also suffered dramatic falls after the pandemic, with a record loss of almost three thousand points from Dow Jones on March 2020. Sectors also changed: tourism and travel were among the most affected industries due to border restrictions, while eCommerce increased. Amazon, for instance, registered record net revenue in 2020 and 2021, trend which finished in 2022.

III. GHANA

1) Introduction and History: Ghana is a country located in the west of the African continent. This country shares borders with Cote d'Ivoire, Burkina Faso, Togo and Gulf of Guinea. It is a country famous for their mines, specially gold mines. Ghana's history can be traced back to 5800 years ago, thanks to archaeologists discoveries (1). In the 9th century Arab merchants wrote and documented Ghana and their trade relationships with the country. This is considered as one of the oldest pieces of evidence of the country (2).

Before the colonization era many kingdoms considered Ghana's soil as the most important in the Asante Empire. Some time later, Ghana was referred to as the "Gold Coast" by Portuguese colonizers in 1471. In the 17th century, Ghana was one of the major slaves exporting countries around the world, due to said colonists. Throughout the 19th century, Western European countries such as Great Britain started focusing on African countries like Ghana, pursuing the colonization and the natural reserves. Finally, in 1957 Ghana gained its independence from Britain and declared the country Ghana, a union between the Gold Coast, Ashanti, the

Northern Territories and British Togoland. Ghana was the first sub-Saharan country to shake off the colonial rules and inspired other countries that struggled to fight for their liberty. Since 1992 Ghana has been under a multi-party democratic model, gaining consideration from other countries as one of the most stable countries of the region. Since December 2024 Ghana's president is John Mahama. It is a diverse country, with 32.1 million people living in Ghana, and 11 different spoken languages. Finally, their main export products are gold, cocoa and oil (recently).

A. Political and Economic Current Situation

As mentioned above, Ghana's democracy, established in 1992, has been praised as a model in Africa. However, the country still faces challenges such as economic crises and political instability. The government as of today maintains a strong monopoly on power, with secularism ensuring a clear distinction between religion and law. Despite being Ghana a rich country in resources like gold, cocoa and oil, it has poor resource distribution development. This has ranked the country 133rd in the 2021 Human Development Index, as nearly 40% of the population is poor, and 49% lived below the poverty line in 2016. Key sectors like gold mining and FDI (Foreign Direct Investment) are dominated by foreign companies, creating challenges in the banking sector as they rely on foreign exchanges for their economic stability.

B. Gold mining and Export Industry

As mentioned previously, gold is the biggest exported mineral, accounting for 48.4% of gross merchandise exports in 2020. However, there are some challenges in the sector, like power reliability or production costs. There are also external factors influencing the gold exports, such as the global gold prices, economic stability in the importing countries, agreements and tariffs or stricter environmental laws. There have also been recent events fluctuating Ghana's gold trade stability, like current wars and sanctions, or events like COVID-19, topic in which we are going to focus on the project.

C. Ghana's International Relations

Ghana is an active country in international organizations such as the UN (United Nations), and ECOWAS (Economic Community of West African States), promoting regional integration. Due to its history with the country, Ghana also has strong ties with the United Kingdom (Ghana is part of the Commonwealth) and United States. There have been rumors about Ghana joining the BRICS+ platform (3). Also mention that Ghana and South Africa, the other country analyzed in this project, share strong trade investment ties, partnership supported by the Bi-National Commission, as the South African Government News Agency stated in 2024 (4). As seen in the figure 1, we find the biggest gold importers from Ghana. We can see that South Africa is in the top 5 biggest importers, however Switzerland is the country that imports the most gold from Ghana (for the sake of the analysis and due to its geographical situation, Switzerland will be considered the European Union).

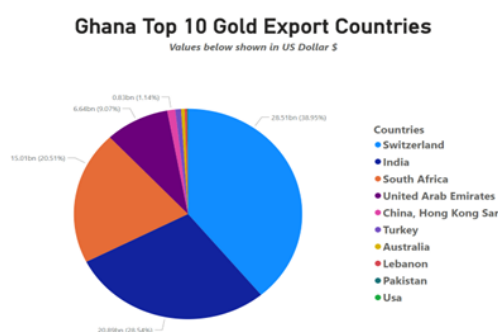


Fig. 1: Ghana's Top 10 Export Countries

D. Impact of COVID-19 on Gold Mining

The 2020 pandemic caused a hit all around the world regarding economic, social and demographic stability. Ghana was no exception to that either. The virus COVID-19 disrupted mining operations due to lockdowns, social distancing and health protocols, which resulted in a decrease on production and exports. This greatly affected Ghana's economy, as they greatly depend on gold trade. However, companies tried to implement safety measures and worker training in order to allow the mining industry to keep forward, but it still delayed the

operations. These delayed operations can be reflected in the export data that we will analyze later. Nevertheless, Ghana has since recovered, with record exports of \$13.92 billion in 2023. The big hit can be seen in the following figure 2, as the year 2019 is the one with the lowest revenue income, and 2020 doesn't even appear.

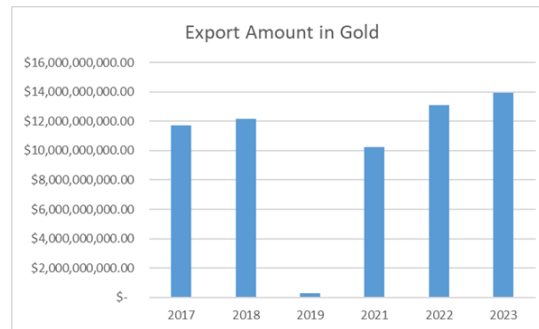


Fig. 2: Ghana's total amount of gold export

IV. SOUTH AFRICA

A. Introduction

South Africa has been the biggest gold producers during several centuries, and it's one of the major exporters nowadays. South Africa's main source of revenue is due to their mineral resource base and, for many decades, its mining industry. This industry supplied many minerals with the most leading contribution of gold mining. South African mines have greatly contributed to urbanization, industrialization and reducing the country's poverty. The biggest problems South Africa faces are unemployment, the slow economic growth and the energy crises, all of them affecting the mining sector's profits. While the government supports mining, their frequent policy changes and the political corruption have led to a deterioration in foreign investments. The strict regulations enforced a lack of operational flexibility and profitability, as the companies are required to meet quotas on Black Economic Empowerment (BEE). Lastly, the changes in the South African Rand affect the export revenue, as gold trade is globally under the USD currency.

B. Impact of the Gold Trade on South Africa's Economy

Gold remains as one of the largest foreign exchange income for South Africa, bringing in billions of rands per year. It also offers direct and indirect employment opportunities, as mining contributes so socio-economic activities. However, the job cuts announced over recent years have brought apprehensions to the topic.

C. Internal and External Factors Regarding South Africa's Mining Industry

South Africa's biggest exported mineral is gold, with the most important gold mine reserves in the Witwatersrand Basin. However, these mines are becoming deeper and more costly to reach. Disruptions are also common due to high labor and unionized strikes. The declining productivity in these mines are due to old infrastructure and inefficiency in operations. Finally, the technological constraints such as automation and advanced mining technologies are being slowly adopted, with safety issues and mine accidents being a huge challenge to the industry.

The internal factors regarding South Africa's gold trade have a big influence on revenues, however external factors also greatly affect the industry's stability. An increased global demand for gold during economic uncertainties is beneficial to South Africa's gold export revenues. The major trading partners with South Africa would include China, India and other emerging markets, as well as their bilateral relation with Ghana. In spite of their bilateral relationship, Ghana remains as a big competitor to South Africa. As South Africa belongs to the BRICS countries, the trade relationships and sanctions influence export trends. Finally, the industry has also faced pressure from the Environmental organizations, with pressure on ecologically friendly practices.

V. GOLD TRADE WITHIN THE BRICS COUNTRIES

As South Africa belongs to the BRICS+ countries, the trade agreements within the bloc benefits the country, while Ghana is in competition with it in various global markets. However, China is a key partner to Ghana, investing in the mining infrastructure, which is in direct competition with South Africa. Even though Ghana is

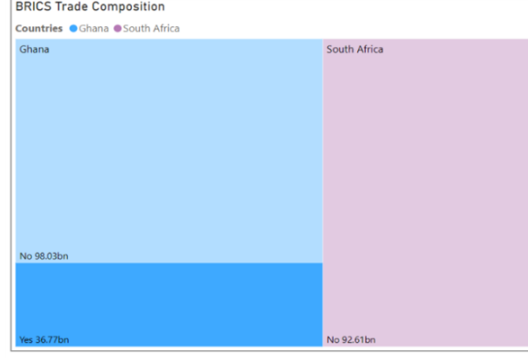


Fig. 3: BRICS Trade Composition

aligning more with BRICS countries, South Africa is still the biggest gold export within these countries, as we can see in the figure 3.

In the tree map Fig. 3 we can see Ghana's "Yes" section, accounting for participation in trade, summing up to a value of \$36.77 billion, while the "No" section representing non-participation, accounts for \$92.61 billion. On the other hand, South Africa's trade with the BRICS nations reflects higher integration, while the \$92.61 billion of total trade under "No," but the absence of a "Yes" section suggests that there are still some possible loopholes within particular trade categories. This graph indicates the balance between established and growing players within the network by highlighting how Ghana's emerging role in BRICS-related trade balances off the dominant position held by South Africa. While mining is an area attracting huge foreign investment, these have fallen in recent years because of policy uncertainties and increased costs of operations. As we can see in the pie chart below in Fig. 4, which shows the contribution of South Africa's gold export from 2017 to 2023, the last year accounts for 27.72% of total exports, reflecting a great recovery and increment in the gold trade. 2019 and 2021 account for the lowest contributions, implying that COVID-19 must have had a big influence in gold trade in this country.

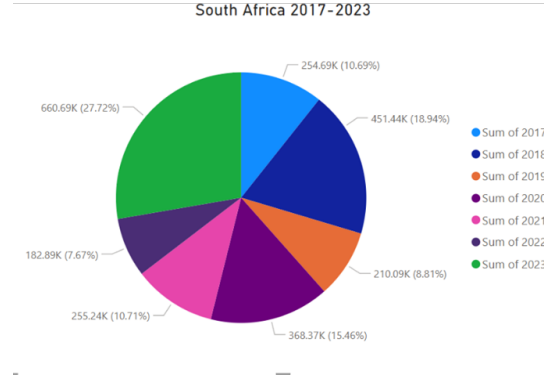


Fig. 4: South Africa's revenue 2017-2023

VI. METHODOLOGY

A. Data Collection

The dataset for this project was sourced from the trade records covering the gold trade between Ghana and South Africa from 2017 to 2023, provided by United Nations Comtrade(5). The data includes any kind of gold export from any of those countries to: countries within the European Union, countries within the BRICS union, USA, UK and Switzerland.

B. Data Cleaning

The raw data required preprocessing before starting the analysis. The steps taken were performed in Python, and they were the following:

- 1) Standardization of column names: all converted to lowercase and whitespaces and special characters replaced with underscores.

- 2) Handling missing and invalid data (rows with missing values in critical columns such as qty were removed, as well as those where qty was zero).
- 3) Duplicate rows were identified and removed to ensure accuracy in the analysis.
- 4) Outlier handling: the numeric rows were examined for outliers. Values above the 99% percentile were capped to the upper limit using the *clip()* method.
- 5) Formatting categorical data: categorical columns like *reporterdesc* and *partnerdesc* were cleaned to ensure consistency (strings got rid of extra whitespace and the capitalization was standardized).

C. Data Analysis

The cleaned data was grouped and analyzed to find patterns. This analysis includes grouped data by reporter (Ghana or South Africa) and afterwards by commodity code (cmdcode). For each of the groups, statistical values such as mean, median, std deviation etc were created using the *describe()* method from the *pandas* library. These summaries were used for posterior insights. Finally, the data was grouped by trade partners (BRICS, EU) to understand the relationship of the exporters with the country blocs. Switzerland was included in the European Union bloc.

D. Visualization

The visualization part includes both PowerBI and python libraries, were histograms, line charts, bar graphs and geographic maps were created to ensure a better vision of the results. These were created using the python library *matplotlib* and PowerBI dashboards.

VII. RESULTS: COVID-19 IMPACT ON SOUTH AFRICA'S AND GHANA'S GOLD TRADE

The COVID-19 pandemic disrupted mining because of lockdown and labor shortages, hence reducing production in 2020. Gold export values, rose during 2020, declining in the post-recovery period (end of 2020, during 2021). Government aimed during this year to cushion the pandemic's negative impact on the economy.

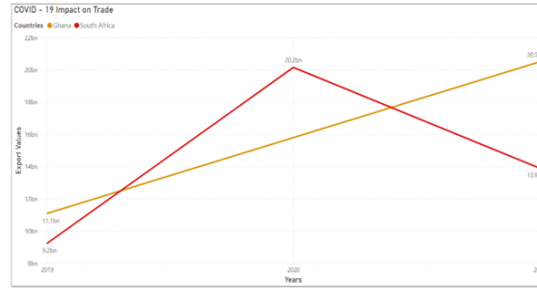


Fig. 5: COVID-19 Impact on Gold Trade

In the figure 5 we can see a line chart comparing the impact of COVID-19 on gold trade between South Africa and Ghana since 2019 to 2021. In 2020, gold values rose in South Africa to about \$20.2 billion due to increased global demand for gold during the pandemic. However, the exports fell back drastically in 2021 to \$13.9 billion, as global markets began to stabilize. Ghana, however, increased export values from \$11.5 billion in 2019 to \$20.5 billion in 2021, proving to be resilient in gold trading. Both countries have had opposite trends during the pandemic era.

The following Figure 6 shows an overall image of the trade from 2017 to 2023 of both countries, giving a clear image of how Ghana recovered greatly while South Africa's gold exports, despite taking a few hits and increasing in 2020, has remained practically the same.

To deepen more into the COVID-19 impact on the trade, we'll divide it into pre-pandemic, during pandemic, and post-pandemic visuals and analysis.

A. Trade Insights and Trends

Ghana's gold exports, as we can see in the Table I, show consistent growth, with Switzerland emerging as the top importer, since 2019. Before that, in 2017 and 2018 the data shows that India was the top importer, in 2017 and as second main importer, after United Arab Emirates. However, in 2019 Switzerland became the biggest importer, maintaining its place until 2023. South Africa has always been in the top gold importers from Ghana, however until 2021 it wasn't stated as top 2 gold importers from said country.

On the other hand, South Africa's export rate has remained steady, with consistent exports to broader regions (Areas, Nes) and minimal activity with neighboring countries such as Botswana and Namibia, as shown in the Figure 7. This highlights South Africa's reliance on global trade networks rather than regional trade.

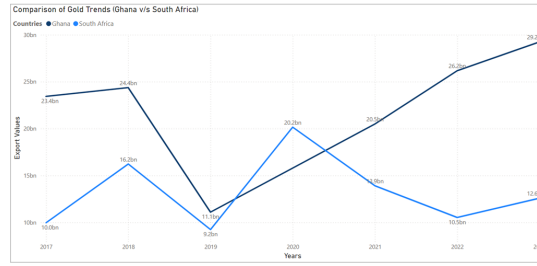


Fig. 6: South Africa's and Ghana's gold trade 2017-2023

TABLE I: Ghana's biggest gold export countries 2017-2023

Year	Status	Country	Gold exported (Kg)
2017	Biggest Export	India	2,010,010
2018	Biggest Export	UAE	1,760,925
	Second Biggest	India	1,470,458
2019	Biggest Export	Switzerland	1,250,843
2021	Biggest Export	Switzerland	2,640,338
	Second Biggest	South Africa	1,351,348
2022	Biggest Export	Switzerland	3,519,203
	Second Biggest	South Africa	2,414,805
2023	Biggest Export	Switzerland	3,427,173
	Second Biggest	South Africa	1,901,107

B. Gold trade 2017-2023

COVID-19 caused significant disruptions in mining operations, including lockdowns, supply chain interruptions and safety protocols. These issues reduced production and delayed exports in both Ghana and South Africa. Ghana's recovery was quicker. In the Figures 8, 10 and 11 we can see that before, during and after the pandemic, the improvement and even higher results as before covid is noticeable. Furthermore, the policies and the investments showed wonderful results as right after the pandemic, the country's gold trade increased back to where it was.

These figures break down Ghana's trade relationships with BRICS, the EU, the UK, and the USA, focusing on pre-covid (2017-2019), during the pandemic (2020-2021) and post-covid (2022-2023).

1) Pre-Covid:

2) *Pre-Covid*: Ghana's gold trade with BRICS countries was relatively stable and moderate in comparison to trade with EU, UK and USA. However, EU, UK and USA were dominant trading country, though less than the BRICS countries. This may be related to China's investment in Ghana's gold trade for the last years. Also, it is important to mention that Ghana's trade relation with other countries has remained minimal, indicating that the focus is on BRICS and traditional Western trading blocs.

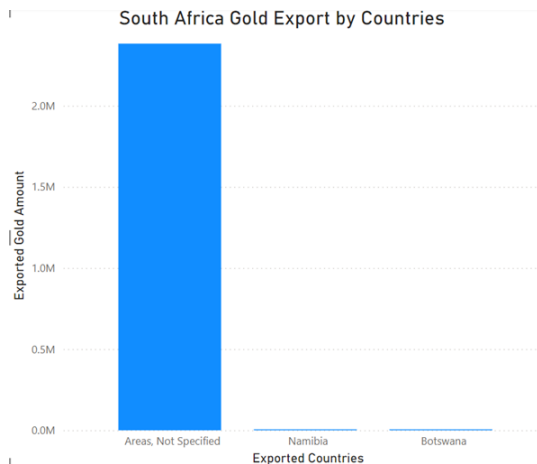


Fig. 7: South Africa's export trade 2017-2023

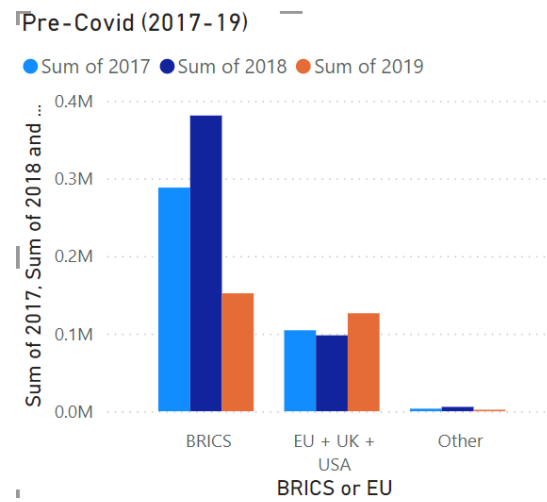


Fig. 8: Ghana's gold trade before the pandemic

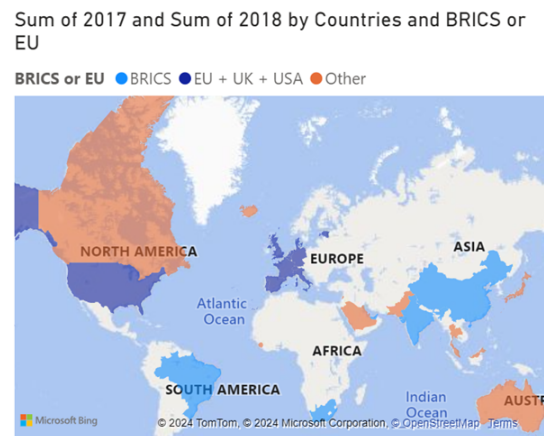


Fig. 9: World map showing Ghana's biggest gold importers pre-pandemic

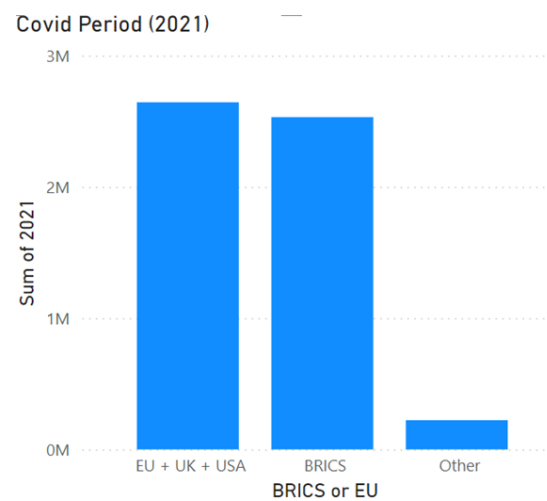


Fig. 10: Ghana's gold trade during the pandemic

3) *During COVID*: Ghana's gold trade with BRICS appears to have increased during the pandemic, almost doubling the gold imports as before COVID. However, the biggest change were EU, UK and USA. These countries and blocs have increased significantly post pandemic, showcasing a bigger dependency on Ghana's gold during the financial instability. This could be due to COVID-related lockdowns, restricted movement of goods and more need of higher gold reserves.

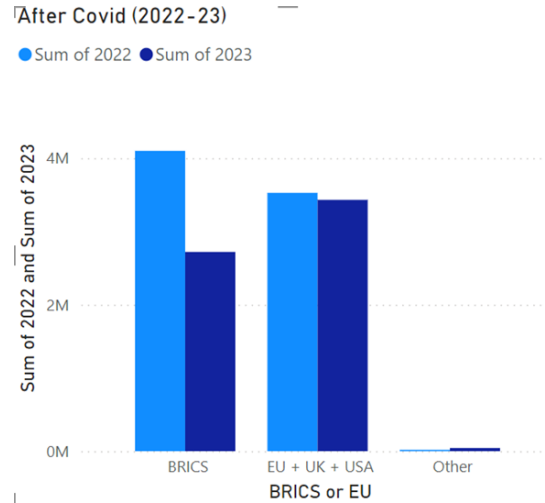


Fig. 11: Ghana's gold trade after the pandemic

4) *Post-Covid*: Finally, we can see that the post-COVID period shows strong trade ties between Ghana and BRICS, as well as EU, UK and USA. It is noticeable that China's involvement in Ghana's infrastructure has likely solidified BRICS as a reliable partner for Ghana's gold exports, threatening South Africa's gold trading territory. Finally, it is important to mention Ghana's gold trade strategy to diversify their resources, balancing relationships with traditional Western partners and BRICS economies to minimize risks associated with over-reliance on certain blocs of countries.

C. Export trade between Ghana and South Africa

As two of the biggest gold exporters in the world, their competition is as big as their bilateral trading relationship. In order to compare the two of them, we can see the Figure 12, where it shows the exports from Ghana to South Africa. In this figure it shows what has been previously stated, that South Africa has significantly increased their imports from Ghana, becoming the second biggest gold importer from Ghana as of today.

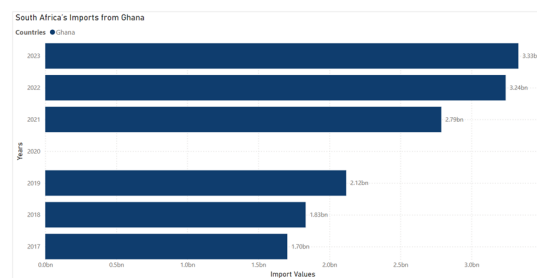


Fig. 12: South Africa's Imports from Ghana

D. Ghana's biggest gold importers

As mentioned, the blocs of countries have been key to Ghana's gold trading sector. However, the Figures 13 and 14 show that Switzerland, considered as one of the richest countries in the world, is the main gold importer (in this analysis we have analyzed it compared to other European countries, even though it is not part of the Union). Even though South Africa and Ghana are competitors, it is no surprise that South Africa imports gold from Ghana. However, as mentioned before in the table I, it is Ghana's biggest importer within the BRICS countries, followed by India. China, despite investing in Ghana's mining infrastructure, does not represent a significant gold importer.

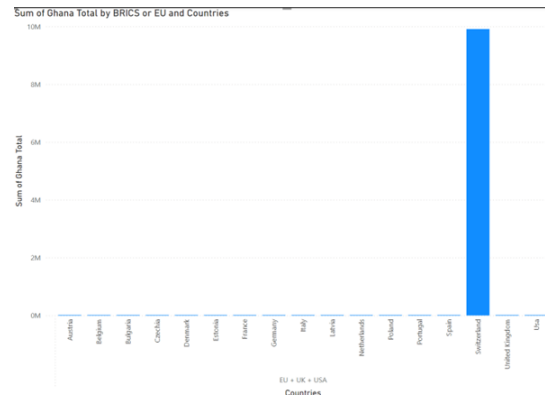


Fig. 13: Ghana's gold trade within the European Union

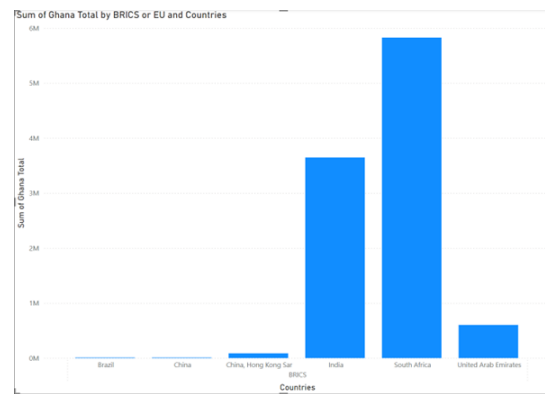


Fig. 14: Ghana's gold trade within the BRICS countries

VIII. CONCLUSION

This analysis has showed the key part that Ghana and South Africa have regarding gold trade. This study has been conducted during a socio-economic instability period such as COVID-19 and the pandemic. Ghana has showed a resilience in their gold exports, associating with rich stable countries like Switzerland and emerging markets like BRICS. On the other hand, South Africa's market showed bigger proof of stability, but not growth. In the future, both nations should prioritize diversification for their international relations (Ghana is on its way), investing in supply chains and strengthen their market relationships to keep growing.

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